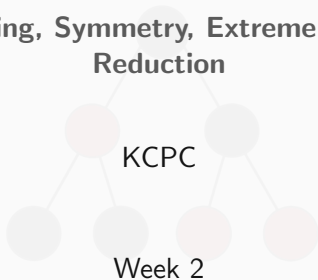


Mathematical Problem Solving II

Wishful Thinking, Symmetry, Extreme Principle, and
Reduction



Overview

- 1 Week 1 Recap
- 2 Heuristic Methods
- 3 Wishful Thinking
- 4 Symmetry
- 5 Extreme Principle
- 6 Reduction



Week 1 Recap



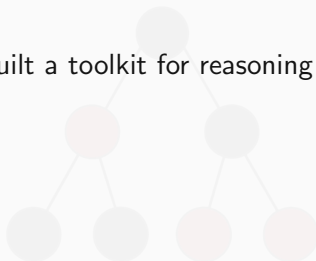
Week 1 Recap

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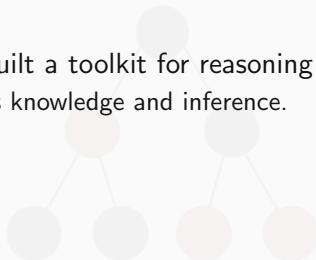
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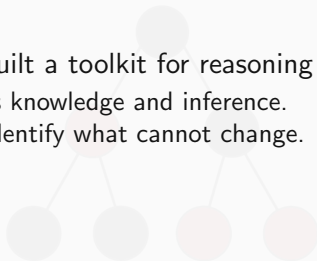
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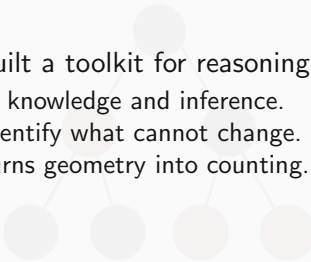


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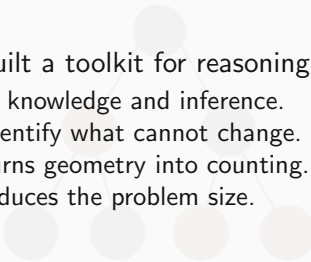
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 - Invariants identify what cannot change.



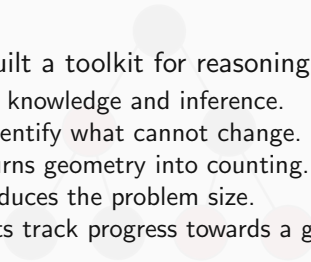
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 - Colouring turns geometry into counting.

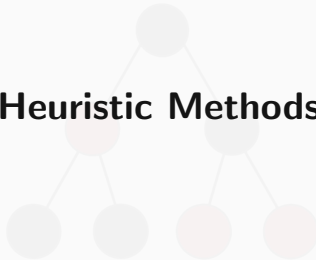
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Heuristic Methods



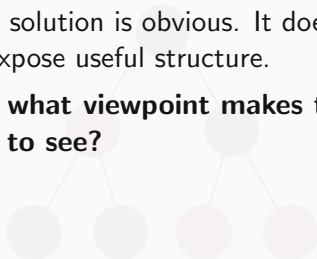
What are Heuristics?



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A **heuristic** is a strategy or rule of thumb that helps you choose what to try when no solution is obvious. It does not guarantee success, but it can expose useful structure.

The key question is: **what viewpoint makes the structure of the problem easier to see?**



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- **Wishful thinking:** imagine a simpler or nicer form.
- **Symmetry:** pair objects or equivalent cases.
- **Extreme principle:** inspect a boundary object.
- **Reduction:** transform the problem into a familiar one.



Wishful Thinking

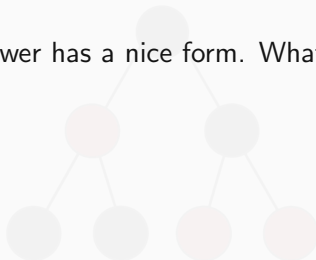
Wishful Thinking

Wishful Thinking: Simplify Until Solvable



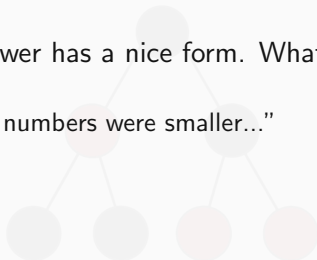
Wishful Thinking: Simplify Until Solvable

→ Assume the answer has a nice form. What would make this easy?

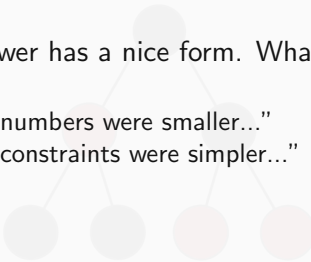


Wishful Thinking: Simplify Until Solvable

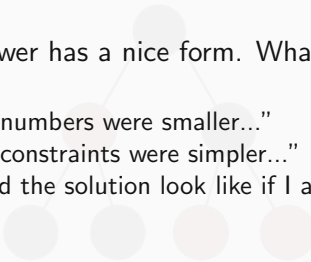
- Assume the answer has a nice form. What would make this easy?
- “If only the numbers were smaller...”



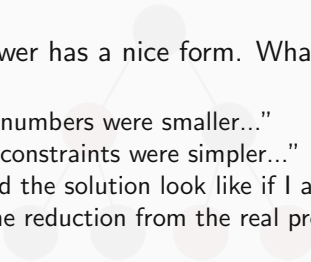
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 - “If only the constraints were simpler...”

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 - “If only the numbers were smaller...”
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 - “What would the solution look like if I already knew it?”

Wishful Thinking: Simplify Until Solvable

- 
- Assume the answer has a nice form. What would make this easy?
 - “If only the numbers were smaller...”
 - “If only the constraints were simpler...”
 - “What would the solution look like if I already knew it?”
 - Then find the reduction from the real problem to the simplified one.

Worked Example: Wishful Thinking – Sum of Squares

Construct infinitely many integer solutions to $x^2 + y^2 = 2z^2$ with $x, y, z > 0$.

Wishful thinking: what if $x + y$ and $x - y$ had a nice structure?

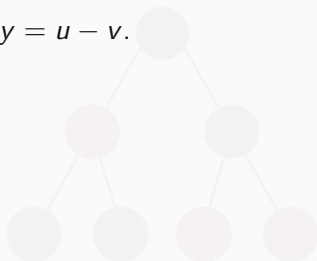
Wishful Thinking

Solution: Sum Of Squares



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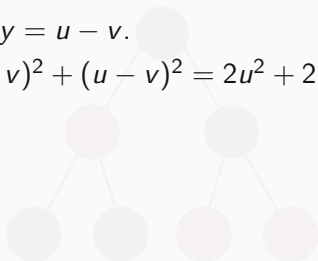
→ Set $x = u + v$, $y = u - v$.



Solution: Sum Of Squares

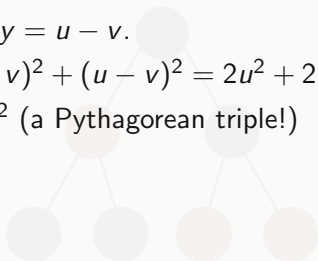
→ Set $x = u + v$, $y = u - v$.

→ $x^2 + y^2 = (u + v)^2 + (u - v)^2 = 2u^2 + 2v^2 = 2z^2$.

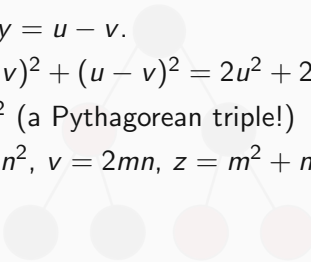


Solution: Sum Of Squares

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- $x^2 + y^2 = (u + v)^2 + (u - v)^2 = 2u^2 + 2v^2 = 2z^2$.
- So $u^2 + v^2 = z^2$ (a Pythagorean triple!)



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 - Take $u = m^2 - n^2$, $v = 2mn$, $z = m^2 + n^2$.

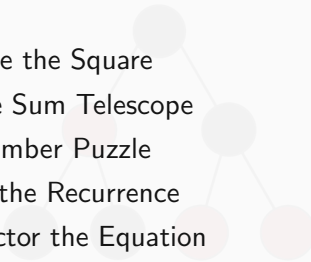
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- Take $u = m^2 - n^2$, $v = 2mn$, $z = m^2 + n^2$.
- Then $x = (m^2 - n^2) + 2mn$, $y = (m^2 - n^2) - 2mn$.

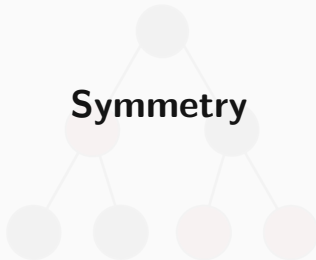
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- Then $x = (m^2 - n^2) + 2mn$, $y = (m^2 - n^2) - 2mn$.
- For example, fixing $n = 1$ and taking $m \geq 3$ gives infinitely many positive solutions.

Student Exercises: Wishful Thinking

- 
- **Basic:** Complete the Square
 - **Easy:** Make the Sum Telescope
 - **Medium:** A Number Puzzle
 - **Hard:** Improve the Recurrence
 - **Very Hard:** Factor the Equation

Symmetry

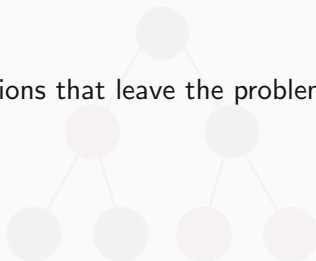


Symmetry: Exploit Invariance Under Transformations



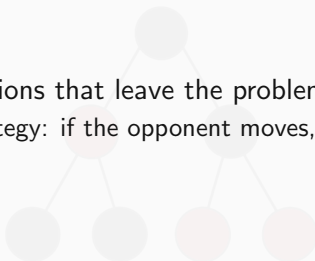
Symmetry: Exploit Invariance Under Transformations

→ Look for operations that leave the problem unchanged.



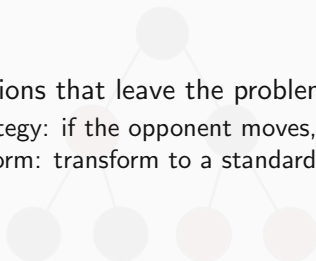
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- Look for operations that leave the problem unchanged.
- Pairing strategy: if the opponent moves, reply symmetrically.

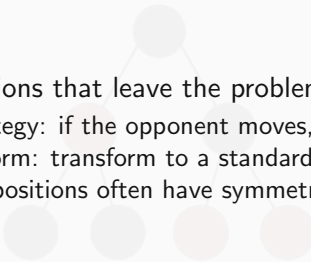


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- Look for operations that leave the problem unchanged.
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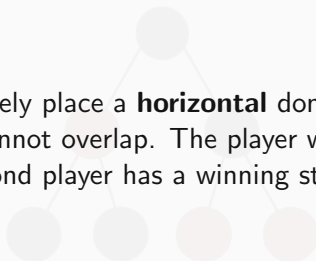


Symmetry: Exploit Invariance Under Transformations

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- Look for operations that leave the problem unchanged.
 - Pairing strategy: if the opponent moves, reply symmetrically.
 - Canonical form: transform to a standard representative.
 - Symmetric positions often have symmetric optimal moves.

Worked Example: Symmetry – Pairing Game

Two players alternately place a **horizontal** domino on a 4×4 board. Dominoes cannot overlap. The player who cannot move loses. Show the second player has a winning strategy.



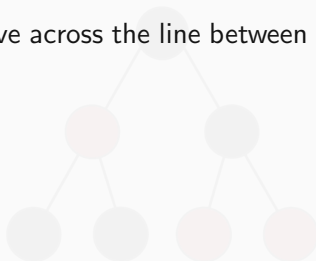
Symmetry

Solution: Pairing Game



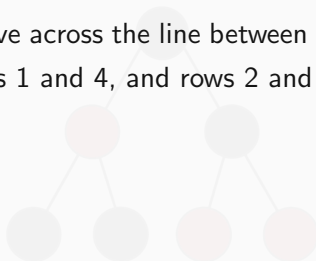
Solution: Pairing Game

→ Mirror each move across the line between the middle two rows.



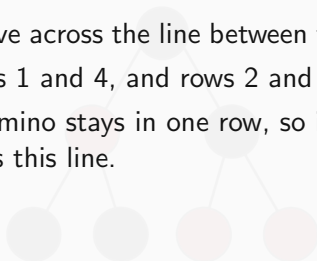
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- Mirror each move across the line between the middle two rows.
- This swaps rows 1 and 4, and rows 2 and 3.

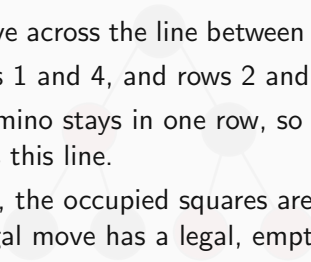


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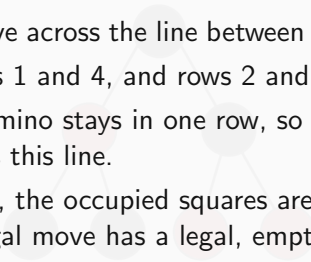
- Mirror each move across the line between the middle two rows.
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- A horizontal domino stays in one row, so it never overlaps its reflection across this line.



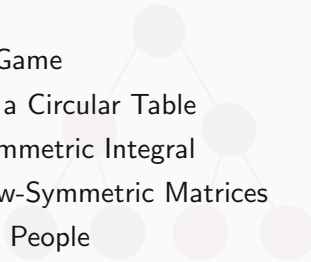
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 - After each reply, the occupied squares are symmetric; the first player's next legal move has a legal, empty mirror.

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 - A horizontal domino stays in one row, so it never overlaps its reflection across this line.
 - After each reply, the occupied squares are symmetric; the first player's next legal move has a legal, empty mirror.
 - The second player never runs out of moves first, so wins.

Student Exercises: Symmetry

- 
- **Basic:** Pairing Game
 - **Easy:** Coins on a Circular Table
 - **Medium:** A Symmetric Integral
 - **Hard:** Odd Skew-Symmetric Matrices
 - **Very Hard:** Six People

Extreme Principle



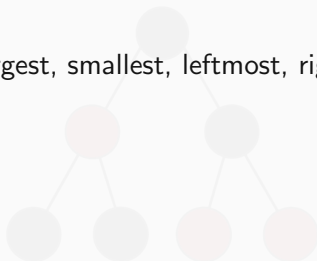
Extreme Principle

Extreme Principle: The Boundary Holds The Answer



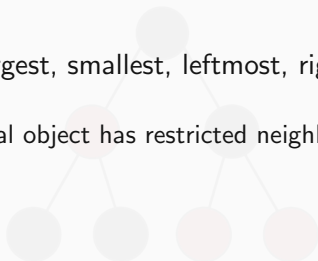
Extreme Principle: The Boundary Holds The Answer

- Consider the largest, smallest, leftmost, rightmost, or most extreme object.



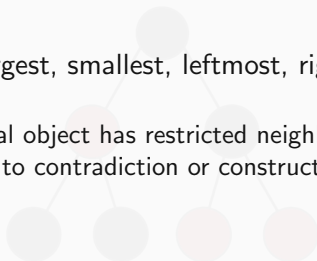
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 - The extremal object has restricted neighbours.

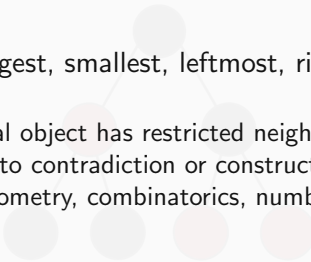


Extreme Principle: The Boundary Holds The Answer

- Consider the largest, smallest, leftmost, rightmost, or most extreme object.
 - The extremal object has restricted neighbours.
 - Often leads to contradiction or constructive greedy choice.



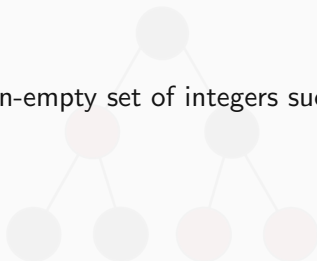
Extreme Principle: The Boundary Holds The Answer

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- Consider the largest, smallest, leftmost, rightmost, or most extreme object.
 - The extremal object has restricted neighbours.
 - Often leads to contradiction or constructive greedy choice.
 - Works in geometry, combinatorics, number theory, and algorithms.

Worked Example: Extreme Principle – A Finite Set

Let S be a finite, non-empty set of integers such that $a + b \in S$ whenever $a, b \in S$.

Prove that $S = \{0\}$.



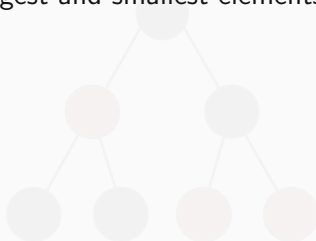
Extreme Principle

Solution: A Finite Set



Solution: A Finite Set

→ Examine the largest and smallest elements of S .



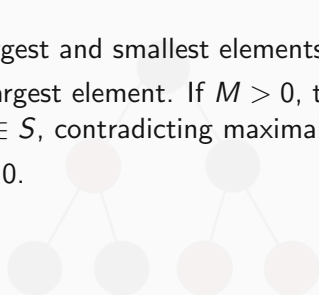
Solution: A Finite Set

- Examine the largest and smallest elements of S .
- Let M be the largest element. If $M > 0$, then $M + M = 2M \in S$, contradicting maximality.



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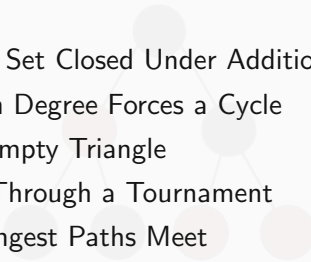
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- Therefore $M \leq 0$.
- Let m be the smallest element. If $m < 0$, then $m + m = 2m \in S$, contradicting minimality.
- Therefore $m \geq 0$.
- Hence $m = M = 0$, so $S = \{0\}$.

Student Exercises: Extreme Principle

- 
- **Basic:** A Finite Set Closed Under Addition
 - **Easy:** Minimum Degree Forces a Cycle
 - **Medium:** An Empty Triangle
 - **Hard:** A Path Through a Tournament
 - **Very Hard:** Longest Paths Meet

Reduction



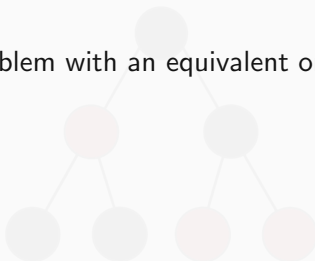
Reduction

Reduction: Change The Representation



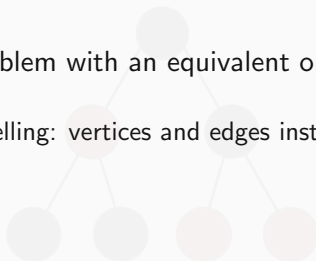
Reduction: Change The Representation

- Replace the problem with an equivalent one that is easier to see.



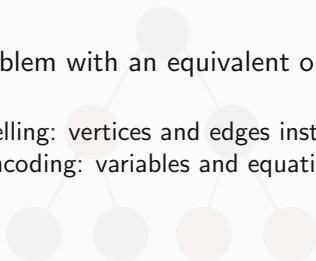
Reduction: Change The Representation

- Replace the problem with an equivalent one that is easier to see.
- Graph modelling: vertices and edges instead of story.

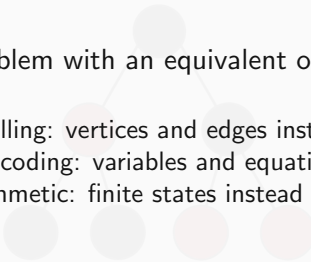


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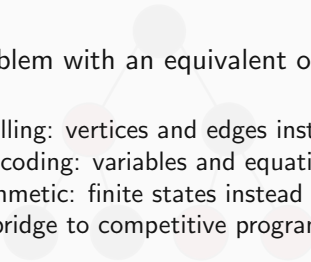
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 - Algebraic encoding: variables and equations instead of logic.



Reduction: Change The Representation

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- Replace the problem with an equivalent one that is easier to see.
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 - Algebraic encoding: variables and equations instead of logic.
 - Modulo arithmetic: finite states instead of infinite.

Reduction: Change The Representation

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- Replace the problem with an equivalent one that is easier to see.
 - Graph modelling: vertices and edges instead of story.
 - Algebraic encoding: variables and equations instead of logic.
 - Modulo arithmetic: finite states instead of infinite.
 - This is the bridge to competitive programming.

Worked Example: Reduction – River Crossing

A farmer must take a wolf, a goat, and a cabbage across a river. The boat carries only the farmer and at most one item.

The wolf cannot be left with the goat, and the goat cannot be left with the cabbage. How can everyone cross safely?

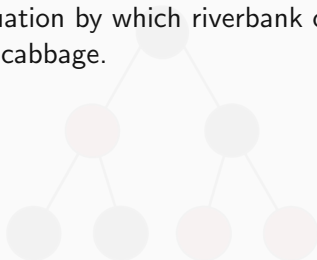
Reduction

Solution: River Crossing



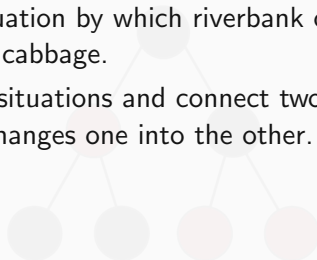
Solution: River Crossing

- Represent a situation by which riverbank contains the farmer, wolf, goat, and cabbage.



Solution: River Crossing

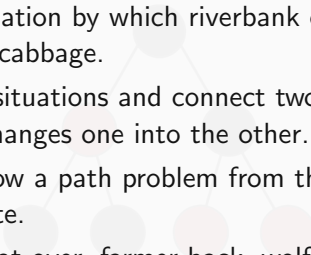
- Represent a situation by which riverbank contains the farmer, wolf, goat, and cabbage.
- Keep only safe situations and connect two when one legal boat crossing changes one into the other.



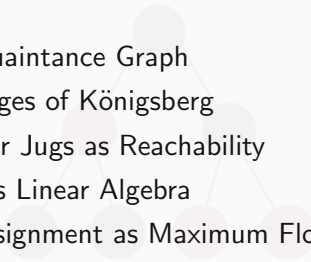
Solution: River Crossing

- Represent a situation by which riverbank contains the farmer, wolf, goat, and cabbage.
- Keep only safe situations and connect two when one legal boat crossing changes one into the other.
- The puzzle is now a path problem from the all-left state to the all-right state.

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- Represent a situation by which riverbank contains the farmer, wolf, goat, and cabbage.
 - Keep only safe situations and connect two when one legal boat crossing changes one into the other.
 - The puzzle is now a path problem from the all-left state to the all-right state.
 - One path is: goat over, farmer back, wolf over, goat back, cabbage over, farmer back, goat over.

Student Exercises: Reduction

- 
- **Basic:** An Acquaintance Graph
 - **Easy:** The Bridges of Königsberg
 - **Medium:** Water Jugs as Reachability
 - **Hard:** Lamps as Linear Algebra
 - **Very Hard:** Assignment as Maximum Flow

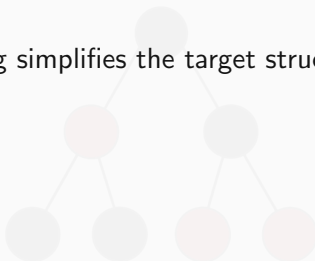
Reduction

Week 2 Complete



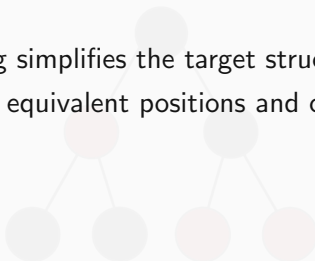
Week 2 Complete

→ Wishful thinking simplifies the target structure.



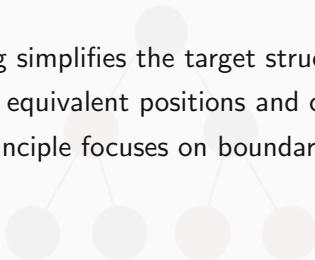
Week 2 Complete

- Wishful thinking simplifies the target structure.
- Symmetry pairs equivalent positions and cases.

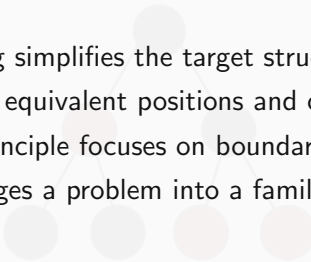


Week 2 Complete

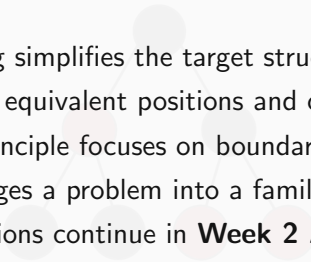
- Wishful thinking simplifies the target structure.
- Symmetry pairs equivalent positions and cases.
- The extreme principle focuses on boundary objects.



Week 2 Complete

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- Wishful thinking simplifies the target structure.
 - Symmetry pairs equivalent positions and cases.
 - The extreme principle focuses on boundary objects.
 - Reduction changes a problem into a familiar representation.

Week 2 Complete

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- Wishful thinking simplifies the target structure.
 - Symmetry pairs equivalent positions and cases.
 - The extreme principle focuses on boundary objects.
 - Reduction changes a problem into a familiar representation.
 - Optional extensions continue in **Week 2 Advanced Slides**.

Thanks for coming!

See ya next week!

I hope you enjoyed the workshop and found some
useful new ways to approach problems.